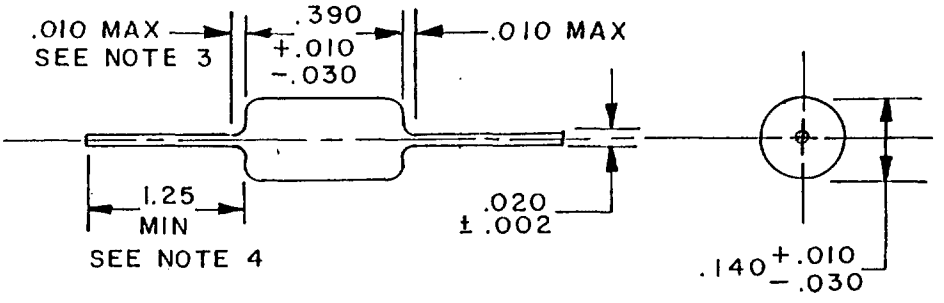
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR83 AND CC83

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-20.

Ⓑ



CIRCUIT DIAGRAM

Inches	mm
.002	0.05
.010	0.25
.020	0.51
.030	0.76
.140	3.56
.390	9.91
1.25	31.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Non-conductive material shall not extend more than .010 inch (0.25 mm) beyond the ends of the capacitor body.
4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Style CCR83 and CC83 capacitors.

Ⓑ denotes changes

AMSC N/A
FSC 5910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Conformal coated.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR83 only): M (1.0%), P (0.1%), R (.01%), or S (.001%).

Thermal shock and voltage conditioning (CCR83 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body Insulation: Test II.

TABLE I. Capacitor characteristics.

Type designation 1/	Rated voltage volts, dc	Nominal capacitance pF	Cap. tol.	Type designation 1/	Rated voltage volts, dc	Nominal capacitance pF	Cap. tol.
CC-83CG151--	200	150	FGJK	CC-83CG821--	100	820	FGJK
CC-83CG181--	200	180	FGJK	CC-83CG102--	100	1000	FGJK
CC-83CG221--	200	220	FGJK	CC-83CG122--	100	1200	FGJK
CC-83CG271--	200	270	FGJK	CC-83CG152--	100	1500	FGJK
CC-83CG331--	200	330	FGJK	CC-83CG182--	100	1800	FGJK
CC-83CG391--	200	390	FGJK	CC-83CG222--	100	2200	FGJK
CC-83CG471--	200	470	FGJK	CC-83CG272--	50	2700	FGJK
CC-83CG561--	200	560	FGJK	CC-83CG332--	50	3300	FGJK
CC-83CG681--	200	680	FGJK	CC-83CG392--	50	3900	FGJK
				CC-83CG472--	50	4700	FGJK
				CC-83CG562--	50	5600	FGJK

1/ Complete type designation will include the following:
1st dash - Symbol "R" (for style CCR83) or dash will be deleted (for style CC83).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR83 only) or dash will be deleted (for style CC83).

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.

Marking: Method I of MIL-STD-1285, as shown in following example:
JCG - "J" brand (ER only) (J); and characteristic (CG).
151 - Capacitance value.
Ⓑ GM8 - Cap. tolerance (G); FR level, CCR83 only (M); and year code (8 for 1988).
ATM - Lot code (A); and trademark (TM).

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 11

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-1607)

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/91492

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T17:07Z
Check the source to verify that this is the current version before use.